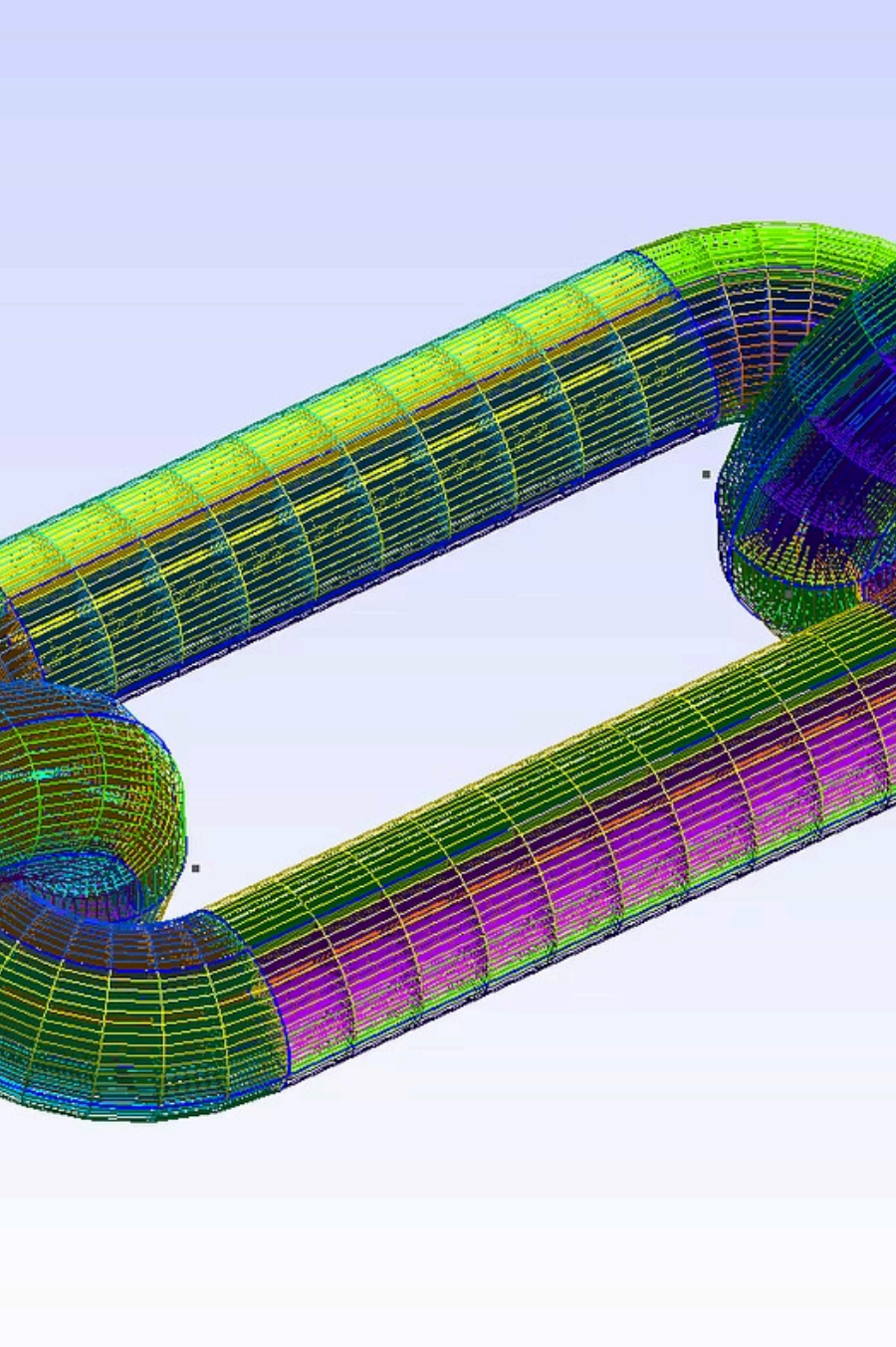




ES 622: Finite Element Methods, Semester Project

Finite Element Analysis of a Mooring Chain Link

Abhijit Venkat, Hari Balaji, Jayeeta Kol

This presentation explores the application of finite element analysis to a mooring chain link, a critical component in offshore oil and gas operations.

Introduction to Mooring Systems and Chain Links

Mooring Systems

Mooring systems are essential for securing offshore structures like platforms, pipelines, and vessels. They consist of anchors, chains, and other components designed to withstand extreme environmental loads.

Chain Links

Chain links are crucial elements within mooring systems, providing strength and flexibility to the chain. Their design and material properties are vital for ensuring system integrity and safety.

Software Used

- Meshing: Gmsh
- Finite element analysis: PrePoMax (with the PARDISO solver)

Objectives of the Study



Stress Analysis

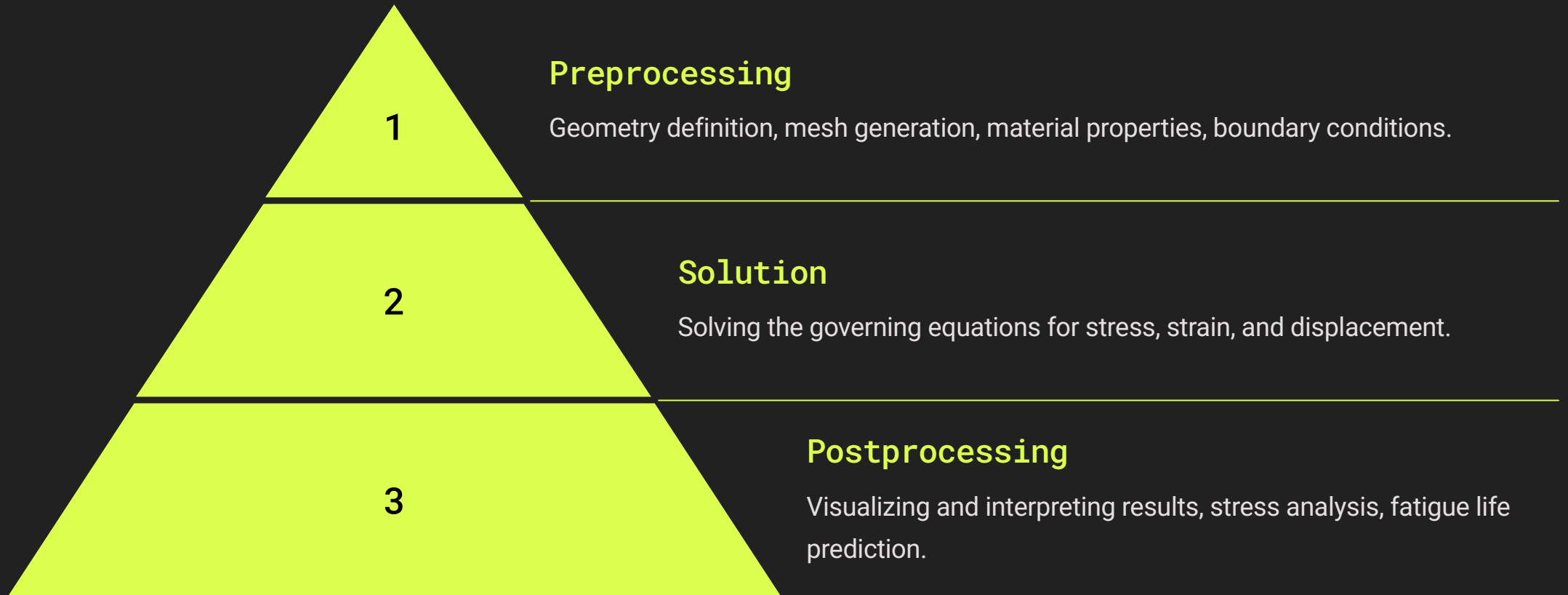
To determine the stress distribution within a chain link under various loading scenarios.



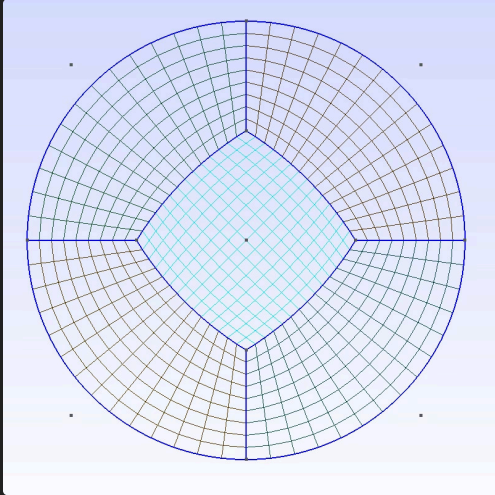
Displacement Analysis

To determine the maximum displacements of the chain under load and find out its deforming behaviour.

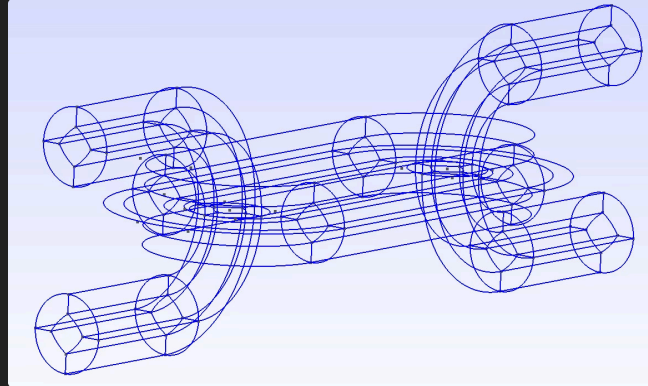
Finite Element Modelling Approach



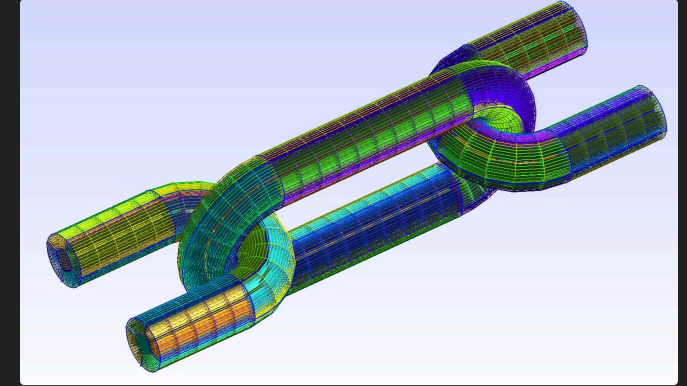
Geometry and Mesh



Meshed Chain cross-section



Completed geometry wireframe



Complete geometry with a structured, 3D mesh.

Mesh Number	Number of Nodes	Number of Quadrangles	Number of Hexahedra
1 (base)	9666	5976	8640
2 (refined)	68704	22950	64800
3 (more refined)	168386	42336	161280
4 (highly refined)	308770	64792	297920

Mesh element used in PrePoMax: C3D10

Material Properties and Boundary Conditions

Material Properties

Density	7800 kg/m ³
Young's Modulus	210 GPa
Poisson's ratio	0.3

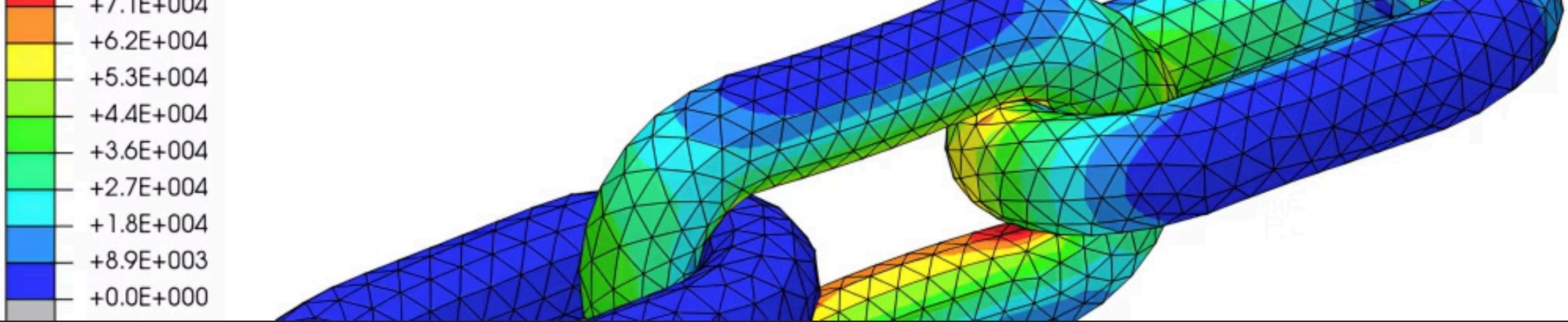
Boundary Conditions

- **Fixed End:** Fixed ($u(0)=0$)
- **Loaded End:** Uniform Pressure BC
Value: **P = 440.0316 MPa**
This value was obtained by dividing the *studless proof load* of a *Grade R4* (equal to **6912 kN [1]**) chain by the area of the cross section on which the load is applied.

Contact Condition

A **hard** contact condition with a friction coefficient of 0.1 was given between the surfaces of the half-links and the full link.

This condition ensures transmission of compressive stresses between the surfaces in contact, and does not permanently enforce a rigid connection like the **tied** condition.



Stress Analysis and Results Interpretation



Stress Distribution

Identifying areas of high stress concentration within the chain link.

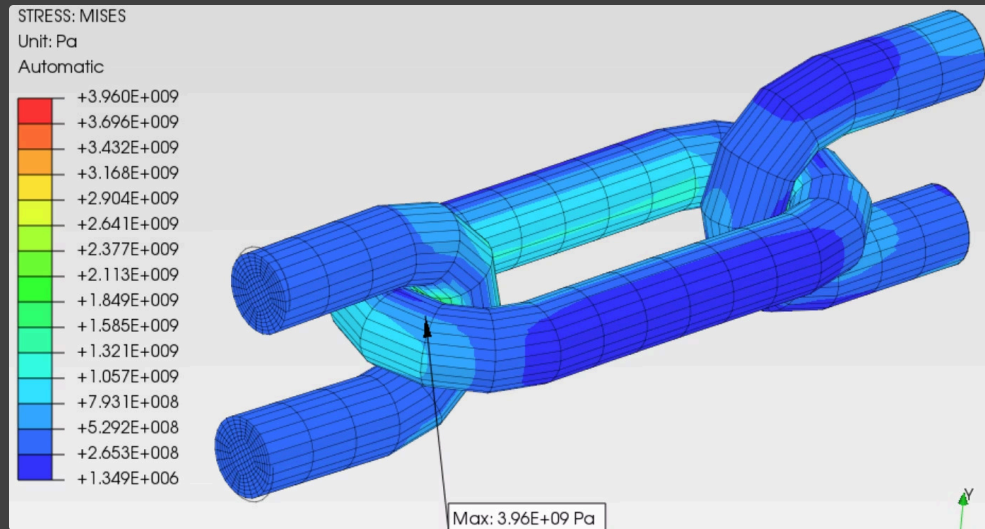


Displacements

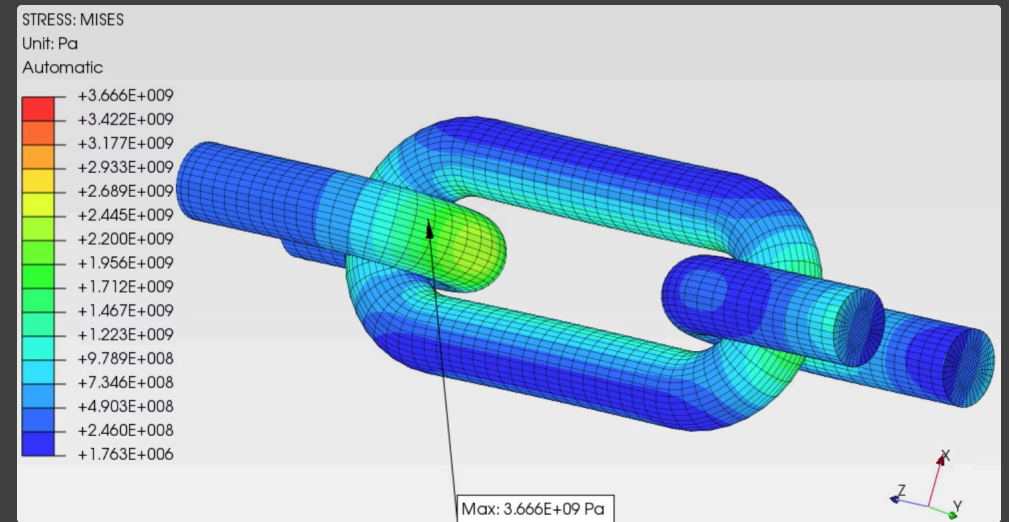
Estimating the number of load cycles the chain link can withstand before failure.

Refinement Level	Max. von Mises Stress (Pa)	Max. x-displacement (m)	Max. y-displacement (m)	Max. z-displacement (m)	Max. total displacement (m)
0 (base)					0.009413
1 (refined)					0.01628
2 (more refined)					0.01653
3 (highly refined)					0.01665

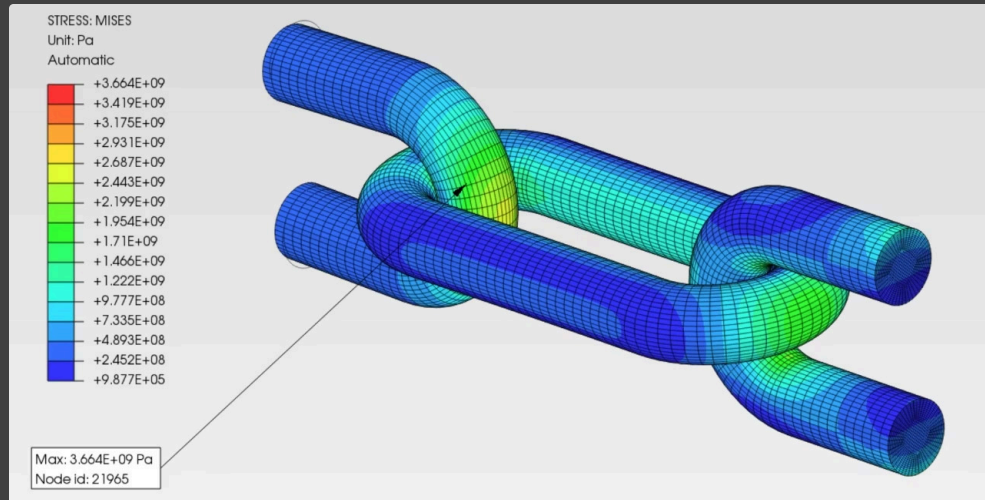
Von Mises Stress Contours



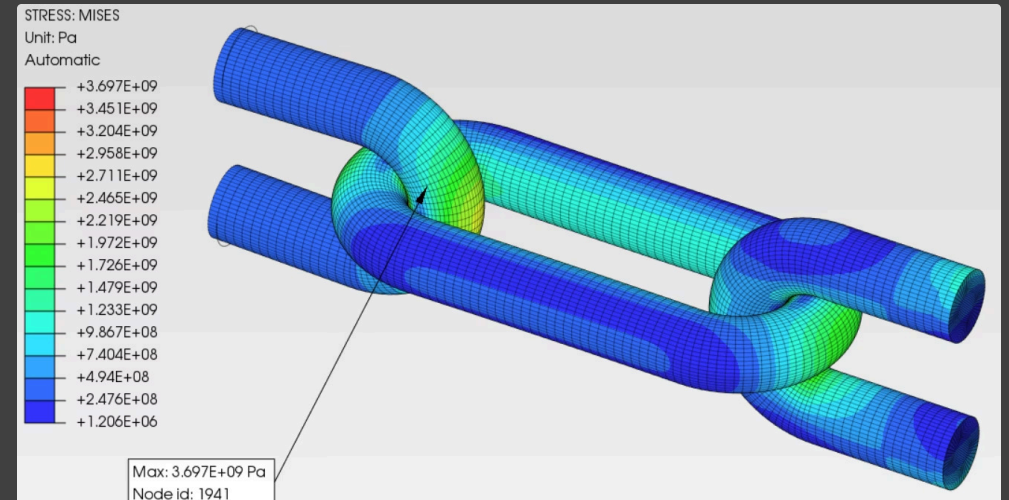
Base Mesh



Refinement Level 1

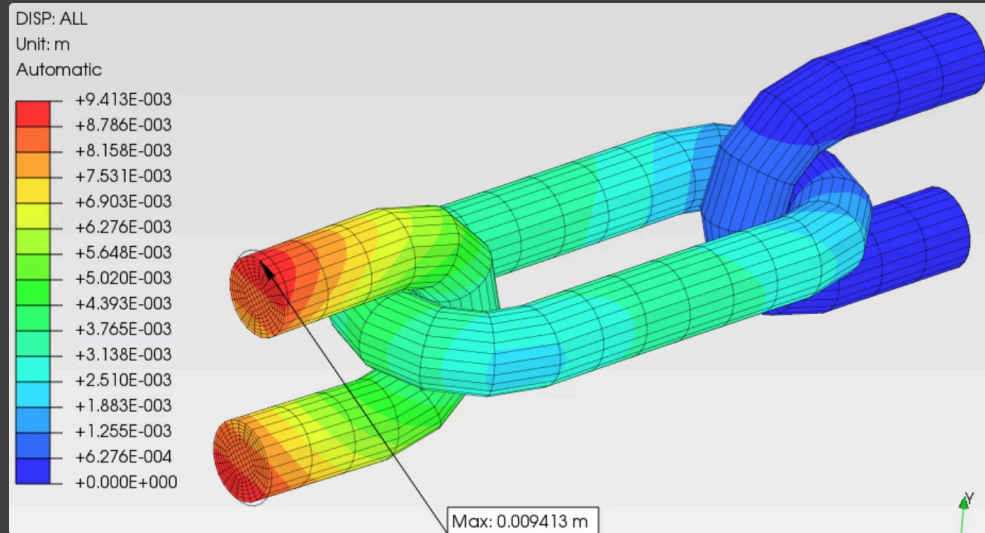


Refinement Level 2

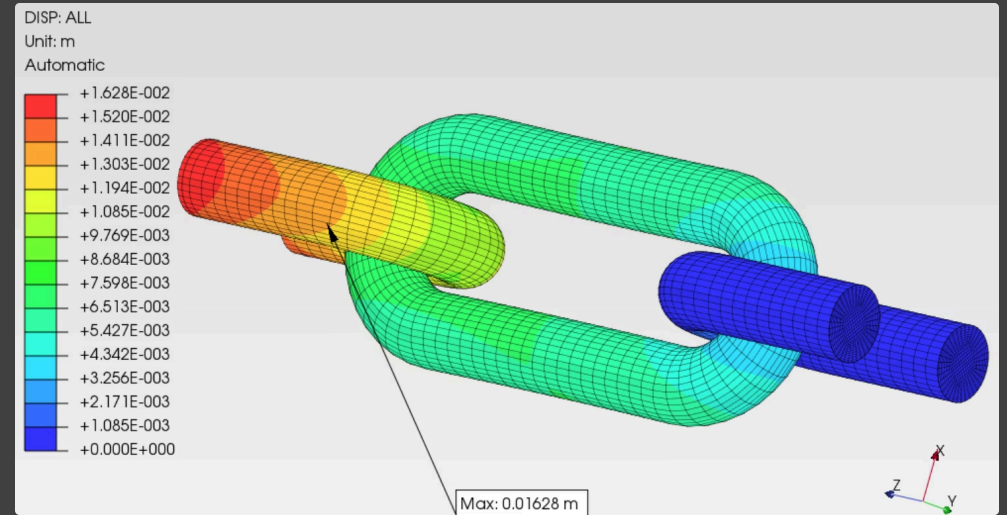


Refinement Level 3

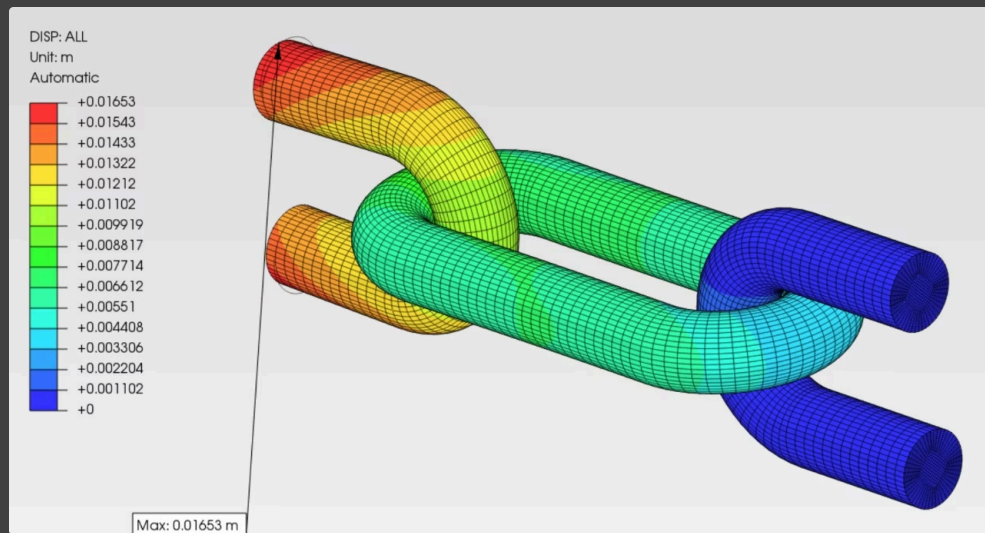
Total Displacement Contours



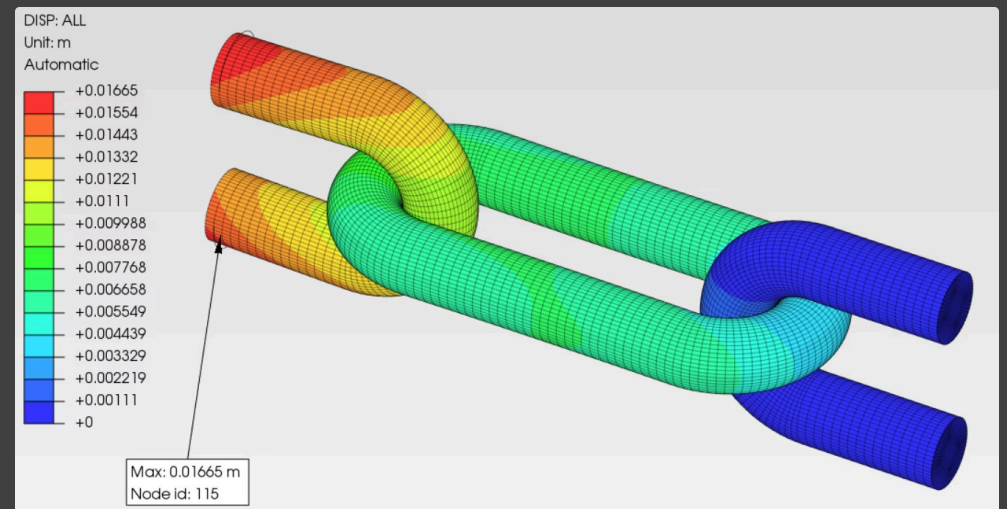
Base Mesh



Refinement Level 1



Refinement Level 2



Refinement Level 3

Convergence

Mesh Refinement

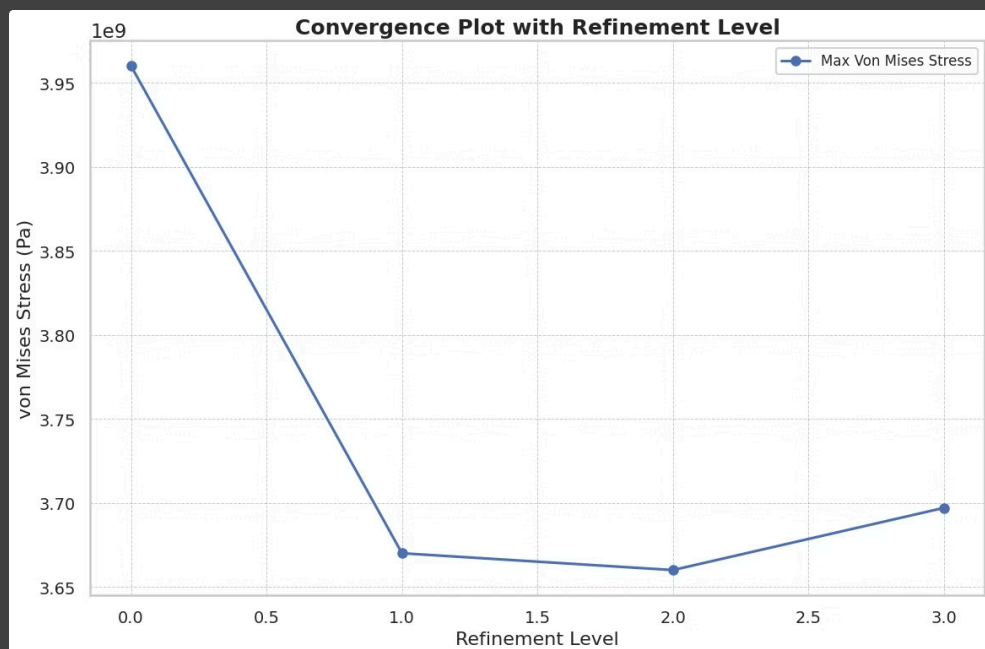
Three levels of mesh refinement were used as mentioned. This was done to ensure mesh independence of the results.

Solution Accuracy

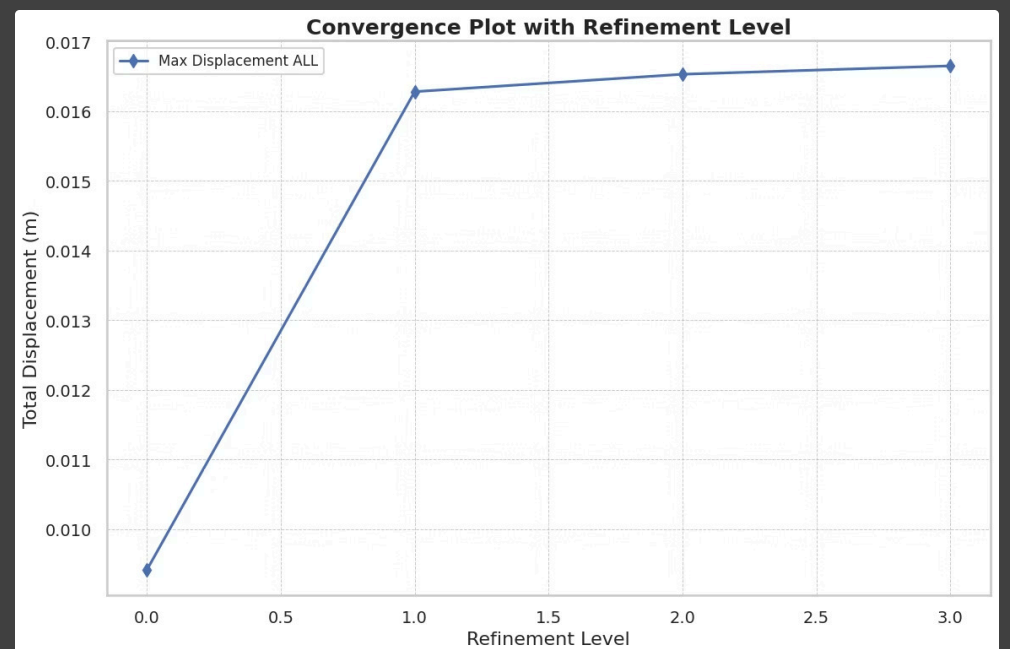
Even though the mesh was refined to quite an extent, the solution did not reach mesh independence. This is because the problem is a contact problem and it has inherent dependence on the number of mesh nodes and/or surfaces in contact with each other.

So, refining the mesh increases the number of nodes and consequently affects the results.

Convergence Plots



von Mises Convergence



Displacement Convergence

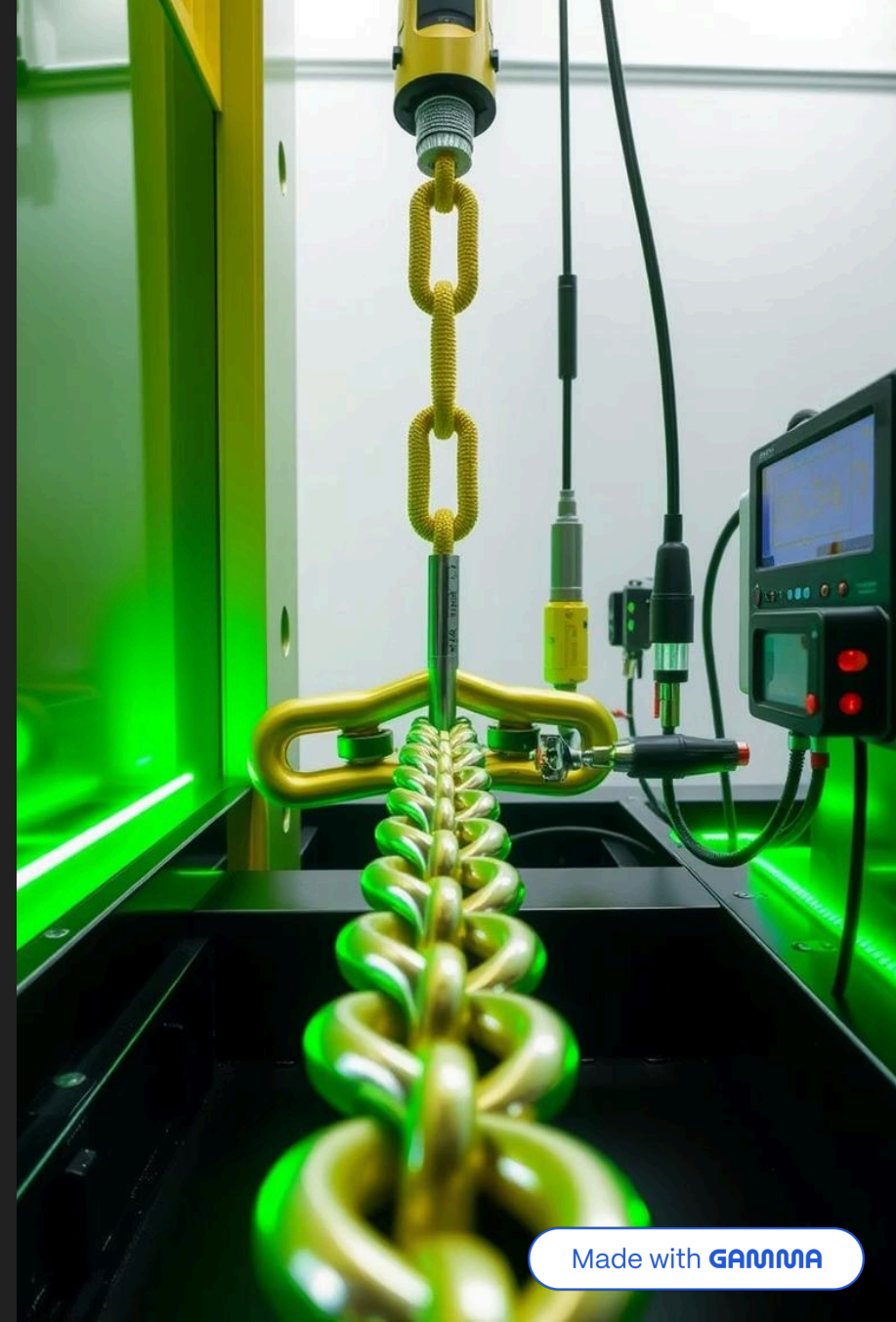
- For the base mesh, the Maximum von Mises stress is higher than the refined cases.
- We can see that the maximum von Mises stress magnitude decreases and the gap between successive refinements gets smaller.
- This indicates that the solution is converging.
- However, it would take several more refinements to achieve complete mesh independence, and some anomalous behaviour can be observed in the third refinement level.

Validation

Since there are no definitive validation cases for our problem, we have to compare our solution to the best possible analogue available for comparison.

Comparison with a simple axial member

Comparing the chain with an axial member of the same cross sectional diameter as that of the chain link will provide significant insights.



Validation

- For a simple axial member, the axial stress is given by $\sigma = F/A$.
- For our case, $F = 6912$ kN, and $A = 2 \times \pi \times R_{cs}^2 = 0.01571$ m²
- Now using the above formula for axial stress, we get $\sigma \approx 440$ MPa.
- Also, in case of pure axial stress, the von Mises stress is equal to the axial stress itself. In our simulation, we expect a similar magnitudes of stress away from the contact points.
- At the contact points, we cannot effectively comment on the magnitudes of stress using this validation case due to the complexity of the contact conditions which prevent this validation from correctly capturing the effects of deformation at the contact points.

Conclusions

1 — Parallel bars of chain-link

The yield strength of high-strength steel used in the manufacture of mooring chains is equal to about 580 MPa [2]. As long as the von Mises stress is lower than the Yield strength of the material, it hasn't undergone plastic deformation and can regain its original shape. As we can see from the contour plots, the von Mises stress magnitude is lower at the parallel bars, which can be compared to our analogous axial force member. This stress is lower than the Yield Strength, and therefore the material hasn't yielded in these locations.

2 — Surfaces in contact

The von Mises stress magnitude is very high near the contact points, indicating that the material has undergone plastic deformation. This also seems logical because the contact points are often the locations at which failure occurs.

3 — Displacements

The displacements also seemed reasonably small and are certainly comparable to physical values, thus solidifying that our solution is on the right track even if it is not completely accurate at the present moment.

References

[1] https://ethz.ch/content/dam/ethz/special-interest/mavt/mechanical-systems/mm-dam/documents/Notes/IntroToFEA_red.pdf

[2] <https://pilotfits.com/products/mooring-chain/>

[3] <https://www.sotra.net/wp-content/uploads/2021/08/CHAIN-MECHANICAL-PROPERTIES.pdf>

AI Disclosure

For the completion of this project, ChatGPT was used to paraphrase some sentences in the report and presentation, and Gamma AI was used to make the template for this presentation.